



MSc Preliminary Project Report

Quantum Local Search for the Max Cut Problem



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1. Introduction

1.1 Problem Context

This project is situated in the field of combinatorial optimisation, which is concerned with selecting the best solution from a large but finite set of discrete possibilities. Many important real-world tasks, including network design, routing, scheduling, and graph partitioning, can be formulated as combinatorial optimisation problems, and these problems are often computationally difficult because the number of possible solutions increases rapidly as the size of the instance grows. Within this domain, the present project focuses on the Max-Cut problem, in which the vertices of a graph are divided into two disjoint sets such that the number or total weight of edges crossing the partition is maximised. Max-Cut is a standard NP-hard benchmark problem and is widely used in optimisation research because it is easy to define, practically relevant, and suitable for testing both classical and quantum methods.

1.2 Motivation

The motivation for this project arises from the limitations of classical local search when applied to difficult combinatorial optimisation problems. Local search methods are attractive because they are simple, flexible, and often effective in finding good approximate solutions within a reasonable computational budget. However, these methods can become trapped in local optima, meaning that no improving move exists within the current neighbourhood even though better global solutions may still be available. This becomes more noticeable in complex optimisation problems, where more effective exploration strategies are needed. Recent work on hybrid quantum optimisation suggests that quantum or quantum-inspired local improvement procedures may help address this limitation by exploring structured neighbourhoods in new ways while remaining compatible with current hardware constraints.

1.3 Project Aim

The main aim of this project is to investigate a hybrid Quantum Local Search framework for the Max-Cut problem and to assess its effectiveness in comparison with a purely classical local search approach.

1.4 Objectives

The objectives of the project are as follows:

- To review the background theory of combinatorial optimisation, Max-Cut, classical local search, and quantum or quantum-inspired optimisation methods.
- To implement a classical local search algorithm for Max-Cut as a baseline method for comparison.

- To design and implement a hybrid QLS-inspired algorithm for Max-Cut using ideas from current quantum optimisation research.
- To test both approaches on benchmark graph instances and evaluate their performance using cut value, convergence behaviour, and computational cost.
- To analyse the strengths and limitations of the hybrid approach and discuss its practical value for near-term optimisation.

1.5 Methodology

The project will follow an experimental and comparative methodology. First, a literature review will be carried out to establish the theoretical background in combinatorial optimisation, local search, quantum annealing, and hybrid Quantum Local Search frameworks. Next, a classical baseline solver for Max-Cut will be implemented in Python using a suitable local search strategy, such as iterative improvement based on vertex flips. A hybrid QLS-style method will then be developed in which local neighbourhoods or subproblems are selected and improved using quantum-inspired optimisation ideas. Both algorithms will be tested on the same benchmark graph instances, and their performance will be compared in terms of solution quality, convergence, and runtime.

From a technical perspective, the implementation will be carried out primarily in Python using standard scientific and graph-processing libraries such as NumPy and NetworkX. There is also potential to make use of quantum programming frameworks such as Qiskit, Cirq, or PennyLane in order to prototype or simulate quantum-inspired local search components, particularly for small subproblems or variational circuit experiments related to Max-Cut. These libraries would be used mainly in simulator mode to support experimentation with hybrid quantum-classical workflows rather than to depend on physical quantum hardware. The main focus of the project, however, will remain on algorithm design, implementation, and empirical evaluation.

1.6 Problem Selection

The Max-Cut problem has been selected as the main case study for this project for several reasons. First, it is a well-known NP-hard problem that provides a meaningful and widely accepted benchmark for evaluating optimisation algorithms. Second, Max-Cut has a natural formulation in terms of graph structures and is frequently used in both classical approximation research and quantum optimisation studies, including work based on QAOA and related hybrid methods. Third, benchmark graph instances can be generated easily in a controlled manner, making Max-Cut a practical and manageable problem for implementation and experimentation within the scope of an MSc project. Although the primary focus

will remain on Max-Cut, the general framework developed in this project may later be extended to other combinatorial optimisation problems such as graph colouring or the travelling salesman problem.

2. Background

2.1 Combinatorial Optimisation

Combinatorial optimisation is a branch of optimisation that focuses on problems where the feasible solutions are drawn from a finite, typically very large, set of discrete configurations. Classic surveys describe it as the study of selecting an optimal solution from a discrete search space defined by combinatorial structures such as graphs, sets, or permutations, with respect to an objective function to be minimised or maximised. Examples include routing, assignment, and scheduling problems that underpin applications in transportation, production planning, network design, and many other areas of operations research and computer science.

A common formal view is that a combinatorial optimisation problem consists of a finite set of feasible solutions and an objective function assigning a numerical value to each solution. Although the feasible set is finite, its size often grows exponentially with the size of the input, which quickly renders exhaustive search computationally infeasible on all but the smallest instances. Many central problems in this area, such as travelling salesman, various scheduling problems, and numerous graph problems, have been shown to be NP-hard, meaning that no polynomial-time algorithm is known that can always find an optimal solution. Consequently, research has developed a wide spectrum of algorithmic approaches, ranging from exact methods to approximation algorithms and heuristic or metaheuristic techniques, to cope with the inherent computational difficulty.

Within this broad class, graph-based optimisation problems have received particular attention because they capture important applications and exhibit rich structural properties. The Max-Cut problem, which is of central interest in this project, exemplifies this category. Given a weighted or unweighted graph, Max-Cut seeks a partition of the vertex set into two disjoint subsets that maximises the total weight or number of edges crossing between the two parts. It is one of the classic NP-hard problems and has significant applications in areas such as VLSI design and statistical physics. There is substantial theoretical work on Max-Cut, including semidefinite programming-based approximation algorithms and extensive research on heuristic methods for large instances.

2.2 Classical Local Search

Classical local search is one of the most widely used heuristic frameworks for solving hard combinatorial optimisation problems, including many NP-hard graph problems such as Max-Cut. The basic idea is to

begin from an initial feasible solution and iteratively move to a neighbouring solution that improves the objective value, according to a predefined neighbourhood structure on the solution space. Academic books and established studies on heuristic optimisation describe local search as a practical and flexible approach for navigating very large discrete search spaces without exhaustively evaluating every candidate solution.

In formal terms, a local search algorithm works with two key ingredients: a current feasible solution and a neighbourhood operator that defines which alternative solutions can be reached by a small modification. Starting from an initial solution, the algorithm repeatedly examines neighbouring solutions and chooses an improving move, either by selecting the first better neighbour encountered or by selecting the best available improving neighbour. The process continues until no neighbouring solution produces further improvement, at which point the algorithm is said to have reached a local optimum. Variants of stochastic local search introduce randomisation in the choice of starting solution, neighbourhood exploration, or move acceptance, which can improve robustness and help the search avoid poor regions of the solution space.

For the Max-Cut problem, local search is especially natural because the quality of a cut can be updated efficiently after small changes to the partition. A common neighbourhood for Max-Cut is the 1-flip neighbourhood, where a neighbouring solution is obtained by moving a single vertex from one side of the cut to the other. Under this setting, a simple local search procedure repeatedly flips vertices that increase the cut value until no improving flip remains. Although conceptually straightforward, such methods can achieve strong performance in practice and form the basis of more advanced heuristic approaches for Max-Cut.

A substantial body of published work shows that local search can be strengthened through hybrid and metaheuristic extensions. For example, studies on Max-Cut have proposed randomized heuristics that combine Greedy Randomized Adaptive Search Procedure (GRASP), Variable Neighbourhood Search (VNS), and path-relinking, producing near-optimal solutions and, in some benchmark instances, new best known results. These results help explain why local search remains an important baseline in optimisation research despite the availability of more sophisticated exact and approximation methods.

The main advantages of classical local search are its simplicity, scalability, and adaptability. It is often easier to implement than exact optimisation algorithms, can be adapted to many different problem types through suitable neighbourhood definitions, and can produce high-quality approximate solutions within reasonable runtime on large instances. Despite these strengths, classical local search has well-known limitations. The most important is its tendency to become trapped in local optima, where no improving move exists within the chosen neighbourhood even though substantially better solutions may exist elsewhere in the search space. This limitation is one of the main reasons researchers have explored

hybrid strategies that retain the global efficiency of classical local search while using more powerful procedures to optimise selected local regions of the search space.

2.3 Quantum and Quantum-Inspired Optimisation

Quantum optimisation has attracted substantial attention as a possible approach for tackling hard combinatorial optimisation problems that are difficult for classical methods to solve exactly. In broad terms, quantum optimisation methods attempt to exploit quantum mechanical effects, such as superposition and tunnelling, to explore complex search spaces in ways that may differ from conventional classical heuristics. Within the current literature, two of the most prominent approaches are quantum annealing and the Quantum Approximate Optimisation Algorithm (QAOA), both of which have been studied extensively on graph optimisation problems such as Max-Cut.

Quantum annealing is an optimisation approach in which a problem is encoded into an energy function, often in Ising or QUBO form, and the system is evolved so that low-energy states correspond to good candidate solutions. Published work has suggested that quantum annealing may be useful for discrete optimisation problems with rugged energy landscapes. At the same time, researchers have argued that the most practical value of quantum annealers may lie not in performing a single global optimisation pass, but in acting as local or guided search devices embedded within more advanced hybrid workflows.

A second major direction is QAOA, a variational hybrid quantum-classical algorithm originally proposed for combinatorial optimisation problems such as Max-Cut. In QAOA, a parameterised quantum circuit alternates between operators derived from the problem Hamiltonian and a mixing Hamiltonian, while a classical optimiser updates the circuit parameters to improve the expected objective value. Because of this structure, QAOA is often presented as a natural candidate for noisy intermediate-scale quantum devices. Max-Cut has become one of the most common benchmark problems for QAOA, and published studies have examined the algorithm's behaviour on different graph families.

Although both quantum annealing and QAOA are promising research directions, the current literature also highlights important limitations. Realistic simulation studies of QAOA on Max-Cut have suggested that practical quantum speed-up over strong classical solvers is unlikely to appear on very small devices and may require hundreds or even thousands of qubits depending on implementation assumptions. Related work has also shown that shallow QAOA does not always outperform local classical methods. These findings are important because they justify the need for careful and critical evaluation rather than overly optimistic claims about quantum advantage.

2.4 Hybrid Approaches and Quantum Local Search

A major theme in current optimisation research is the development of hybrid methods that combine classical and quantum computation rather than relying on either paradigm alone. This direction has emerged largely because classical algorithms remain highly effective on many practical optimisation tasks, while near-term quantum devices still face important constraints such as limited qubit counts, noise, and restricted circuit depth. As a result, much recent work has focused on architectures in which a classical algorithm manages the overall search process and a quantum or quantum-inspired routine is used selectively for difficult subproblems or structured local improvement steps.

This hybrid perspective is especially relevant for combinatorial optimisation problems, where global exact solution is often infeasible and practical performance depends on the quality of heuristic search strategies. Classical local search provides a natural outer framework because it already decomposes the optimisation process into a sequence of local improvement steps. The key idea behind hybrid local search is therefore to preserve the scalability and simplicity of classical search while delegating carefully chosen neighbourhood or subproblem optimisation tasks to a different computational engine that may explore those local regions more effectively.

Published work by Tomesh, Saleem, and Suchara provides one of the clearest formulations of QLS as a hybrid optimisation framework. In that work, the authors propose a quantum local search algorithm based on the Quantum Alternating Operator Ansatz and apply it to the Maximum Independent Set problem. Rather than attempting to solve the full problem on a quantum device, the method repeatedly identifies local neighbourhoods, optimises those neighbourhoods on a small quantum circuit, and then combines the locally improved solutions into a larger global solution. The reported results show that this strategy can address graph instances with up to 100 nodes while using far fewer qubits than would be needed to represent the full problem directly.

At the same time, the existing literature also makes clear that QLS is not a complete or settled solution. Its performance depends strongly on how local subproblems are selected, how neighbourhoods are represented, and how effectively the quantum subroutine can optimise the selected region. Recent work has attempted to improve the framework through more efficient encodings and qubit-efficient designs. These developments suggest that QLS is a promising but still evolving approach.

2.5 Related Work

Several published studies are relevant to this project because they show how classical, quantum, and hybrid methods have been used for combinatorial optimisation problems such as Max-Cut. The literature does not yet show one clearly best method for all cases, but it does provide useful ideas and evidence for designing and evaluating a hybrid Quantum Local Search approach.

A key paper for this project is *Quantum Local Search with the Quantum Alternating Operator Ansatz* by Tomesh, Saleem, and Suchara. This work presents a hybrid method in which a classical algorithm selects local neighbourhoods and a quantum routine is used to optimise those smaller subproblems. Although the paper studies the Maximum Independent Set problem rather than Max-Cut, it is very important because it introduces the main idea of Quantum Local Search and shows that larger graph problems can be tackled using relatively small quantum resources.

Another important study is *QAOA for Max-Cut requires hundreds of qubits for quantum speed-up* by Guerreschi and Matsuura. This paper reports simulation results suggesting that current small quantum devices are unlikely to outperform strong classical methods on Max-Cut, and that many more qubits may be needed before a clear speed advantage appears. This is useful for the present project because it shows that a fully quantum solution may not yet be practical, which supports the choice of a hybrid approach instead.

Classical Max-Cut heuristics are also an important part of the related work. The paper *Randomized heuristics for the Max-Cut problem* presents local-search-based methods such as GRASP, Variable Neighbourhood Search, and path-relinking, and reports near-optimal solutions on benchmark instances. This shows that classical heuristic methods for Max-Cut are already highly developed, so any hybrid method should be compared against meaningful classical approaches rather than weak baselines.

Overall, the related work suggests three main points. First, Quantum Local Search is a promising hybrid idea for using limited quantum resources on larger optimisation problems. Second, fully quantum methods for Max-Cut do not yet clearly outperform strong classical solvers on current small-scale hardware. Third, because classical Max-Cut heuristics are already strong, the value of a hybrid QLS method must be judged through careful implementation and comparison.

2.6 Critical Discussion

The literature reviewed in the previous sections shows that classical, quantum, and hybrid optimisation methods each have strengths and weaknesses. Classical local search methods are simple, scalable, and already perform very well on many Max-Cut instances, especially when combined with stronger heuristic strategies such as GRASP, VNS, and path-relinking. This means that any new hybrid or quantum method must be compared against strong classical baselines rather than weak ones.

At the same time, classical local search has an important limitation: it can become trapped in local optima. If the neighbourhood is too small, the search may fail to escape poor regions of the solution space, but if the neighbourhood is made much larger, the computational cost increases. This creates a trade-off between efficiency and search quality.

Quantum optimisation methods such as quantum annealing and QAOA are appealing because they offer a different way of exploring hard optimisation landscapes. However, published studies also show that current near-term quantum methods still face major challenges, including limited qubit numbers, noise, and the difficulty of parameter optimisation. In the case of Max-Cut, research has suggested that QAOA may require hundreds of qubits before it can clearly outperform strong classical solvers in practical runtime. This means that fully quantum approaches may not yet be the most realistic option for solving large combinatorial optimisation problems in practice.

Hybrid methods such as Quantum Local Search are therefore attractive because they try to combine the strengths of both sides. In this approach, the classical algorithm manages the overall optimisation process, while the quantum or quantum-inspired routine is used only on smaller local subproblems. This is a practical idea because it reduces the amount of quantum hardware required and allows larger problems to be handled than would be possible with a direct quantum encoding of the full instance. At the same time, the literature suggests that the success of QLS depends heavily on how local neighbourhoods are selected and how well the quantum subroutine performs within those neighbourhoods.

Overall, the literature suggests that hybrid optimisation is currently the most realistic direction for near-term quantum computing in combinatorial optimisation. Classical methods remain strong and reliable, quantum methods remain promising but limited, and hybrid approaches offer a practical middle ground that is worth investigating further.

3. Project Schedule

This section outlines a realistic work plan for the project, taking into account the official deadlines and the fact that the examination period from 1 May to 29 May 2026 will significantly limit the time available for project work.

3.1 Phases and timing

The project is divided into six phases:

1. Phase 1: Preliminary report (up to 30 April 2026)
During this phase, the focus is on defining the project and completing the MSc preliminary project report.
2. Phase 2: Examination period (1 May – 29 May 2026)
This period is dedicated mainly to examinations, so only very limited project work is planned.

3. Phase 3: Literature review and detailed design (30 May – 23 June 2026)
After the exams, the project work restarts with an extended literature review and the detailed design of the algorithms.
4. Phase 4: Implementation (24 June – 21 July 2026)
This is the main coding phase of the project, including implementation of the classical baseline and a first version of the QLS-inspired method.
5. Phase 5: Experiments and analysis (22 July – 1 August 2026)
Once the implementations are stable, the project will focus on experiments and analysis.
6. Phase 6: Final report and presentation (2 August – 6 August 2026)
In the final phase, the focus shifts to documenting and presenting the work.

3.2 Summary of timeline

- Up to 30 April 2026: define the project and submit the preliminary report.
- 1 May – 29 May 2026: exams; project effectively paused.
- 30 May – 23 June 2026: literature review and detailed algorithm design.
- 24 June – 21 July 2026: implementation of classical baseline and QLS-inspired method.
- 22 July – 1 August 2026: experiments and analysis.
- 2 August – 6 August 2026: final report and video preparation, and submission.

This plan is intentionally conservative and acknowledges that the exam period will significantly reduce available project time. By concentrating on the main technical work in late June and July, it aims to keep the workload realistic while still allowing enough time to implement the algorithms, run experiments, and produce a coherent final report and presentation.

4. Expected Outcomes

By the end of the project, the following outcomes are expected:

- **Implementations:** A working Python implementation of a classical local search algorithm for Max-Cut and a first QLS-inspired hybrid version.
- **Experimental results:** A small set of experimental results on benchmark graphs, comparing how well the classical and QLS-inspired methods perform.

- **Understanding and reflection:** A written discussion of what worked well, what did not, and whether the QLS-inspired ideas seem practically useful for Max-Cut at this scale.

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